Yr2DSP-D25

PROJECT CHARTER

Date:	21/10/25

1. SUMMARY	
Project Name	AI and Signal Processing Techniques for Parkinson's Finger Tapping
Project Manager	Ebuka
Project Start – End Dates	13/10/25 – 6/04/26
Project Goal (SMART)	By April 2026, develop and validate an AI-assisted assessment tool that analyses finger tapping movements to support the diagnosis and monitoring of Parkinson's disease. The tool will provide an accuracy of at least 85% in identifying symptom severity compared to standard clinical assessments, helping clinicians objectively evaluate and track progress.

2. SCOPE

Purpose: *To develop a visual computing-based assessment that helps doctors and researchers objectively analyse finger tapping movements in Parkinson's patients. The system will use videos and signal data to measure how smoothly and quickly a patient taps their fingers, providing consistent insights into their motor control.*

Background:

- *Parkinson's Disease is one of the most common neurological disorders globally*
- *Finger tapping is a routine clinical test used to assess motor function*
- *Current assessments are judged by eye, making them subjective and sometimes inconsistent*
- *Using AI, we can make this process more accurate, consistent, and data-driven, providing clearer information relevant to diagnosis and patient monitoring*

Inclusion criteria:

- *Focus on analysing finger tapping movements in Parkinson's patients*
- *Use of AI and signal processing techniques to assess movement speed, amplitude, and rhythm*
- *Classification of symptom severity (normal, moderate, severe)*
- *Use of ethically approved and existing patient data*

Exclusion criteria:

- *No medical devices or hospital system integration will be developed*
- *The focus will not include other Parkinson's symptoms (like speech or walking)*
- *No new patient data will be collected*

3. BENEFITS

- Provides an objective and consistent method for diagnosing and monitoring Parkinson's
- Supports clinicians in treatment planning by measuring how severe a patient's symptoms are over time
- Reduces human error and bias in manual observation
- Contributes to early detection and treatment adjustment
- Advances research in AI-assisted healthcare solutions

4. SUCCESS CRITERIA			
Indicator	**Baseline**	**Target**	**Rationale**
Diagnosis accuracy	Manual observation	≥85%	Measures how accurately the AI matches or exceeds doctors' assessments when identifying Parkinson's
Feature extraction	Manual observation	clinically relevant features	Ensures the system captures meaningful signals used in diagnosis
System performance	Prototype	Reliable and repeatable outputs	Confirms the model provides consistent results suitable for patient monitoring
Clinical usability	N/A	Easy-to-use prototype demo	Demonstrates how clinicians can practically use the tool in diagnosis and tracking

5. KEY MILESTONES & DELIVERABLES				
Deliverables (Quantify – be specific)	**Main Tasks**	**Start**	**End**	**Lead**
Project charter	Create and review the charter	21/10/25	31/10/25	Ryan
Meeting with the client	Present the project charter to the client	Week 7	Week 7	Ebuka
Project initiation	Create and review a project plan	Week 7	Week 7	Ebuka
Background research	Conduct a literature review and identify clinical parameters for movement evaluation	Week 7	Week 9	Team
Data understanding and preprocessing	Explore the finger tapping dataset	Week 9	Week 11	Elvis
Feature extraction development	Extract key movement patterns using simple signal analysis	Week 11	Week 12	Avyandra
AI model development	Train the model to classify symptom severity	Week 13	Week 17	Team
Model evaluation and validation	Test the model accuracy and reliability against clinical benchmarks	Week 17	Week 19	Team
Prototype and usability testing	Develop and test simple user interface for clinicians	Week 19	Week 21	Ryan
Final report and presentation	Compile results and present the findings	Week 21	Week 24	All members

6. OBJECTIVES & KEY RESULTS

Objectives	Key Results
Develop clinically relevant feature extraction	Identify clinically relevant features (tap amplitude, rhythm, velocity, time). Validate their relevance through research
Build AI model for diagnosis and monitoring	Achieve ≥85% accuracy in classifying severity. Demonstrate potential for long-term patient monitoring
Evaluate and validate model performance	Compare results with clinician assessments and evaluate using metrics like precision, recall and f1-score
Support clinical usability	Deliver an easy-to-use prototype that can be used in clinical assessments

7. CONSTRAINTS & ASSUMPTIONS

Constraints	Assumptions
Ethical data use only	Dataset access is approved and secure
Technical complexity	Team has foundational knowledge in AI and signal processing
Limited to 6 months	Team members contribute consistently
Limited processing power	Team and university computers will be sufficient

8. POTENTIAL RISKS & ISSUES

Risk Description	Mitigation	Probability (H, M, L)	Impact (H, M, L)
Dataset may lack diversity or quality	Use data augmentation techniques to make the dataset robust if needed	M	H
Technical challenges in model development	Consistently research and consult the supervisor regularly	M	H
Time constraints	Apply effective task allocation and weekly check-ins of progress	H	H
Model overfitting or misclassification	Use cross-validation and keep model complexity manageable	M	H

9. PROJECT BUDGET/ RESOURCES

Budget: *Full-Time Equivalent (FTE)*

1 person = 3 days/week = 0.6 FTE

4 people = 2.4 FTE

Total Labour Yearly Cost (TLYC) = £72,000

6-month project cost = £36,000

Miscellaneous estimated costs = £200

Total budget: £38,000

Resources:

Software: Python, Matplotlib, OpenCV, TensorFlow, SkLearn, Numpy, VSCode, SciPy
Hardware: Personal laptops
Data: Pre-approved Parkinson's patient dataset

10. PROJECT TEAM		
Role	**Name**	**Responsibility**
Client	Prof. Ramzi	Client representative & Project supervisor
Academic Supervisor	Prof. Kulvinder	Academic and project supervisor
Project Manager	Ebuka	Oversees coordination and delivery
Member	Ryan	Model validation and documentation
Member	Elvis	Research and feature extraction
Member	Avyandra	Model implementation and evaluation
Stakeholder	(Provisional: Clinician)	

11. APPROVALS	
Approved by Project Sponsor: Prof. Kulvinder Panesar – University of Bradford	Date: 13/10/25